Report: *Bert has uncommon sense*, Gessler and Schneider (2021)

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Abstract

In their paper *BERT has Uncommon Sense* [1], Gessler and Schneider (G&S) address the challenge of word sense disambiguation by introducing a novel evaluation method based on a query-by-example formulation. In this framework, the embedding of a query sentence is compared to those of sentences stored in a reference database. Through experiments on two distinct corpora (OntoNotes 5.0 and PDEP), they conclude on the effectiveness of BERT like models for word sense disambiguation, despite high variance results between models. Building on their work, we propose to investigate the robustness of this evaluation method under variations in the embedding layer and the similarity metric employed. Additionally, we extend the framework to the another language by applying it to French models and corpora.

1 Paper presentation

1.1 Introduction

With the development of foundation models trained on massive datasets, such as BERT (encoder-based) and GPT-2 (decoder-based), a new area of research has emerged focused on understanding how these models handle polysemy. In particular, the question of how these models embed rare or ambiguous words remains an active topic of research. Classical approaches, such as the one proposed by [2], rely on a k-nearest neighbors (kNN) classifier, where the predicted sense of a word corresponds to the most frequent sense among its k nearest neighbors in embedding space. In [1] authors try to relax the metric used in [2], by making it less stringent and rewarding partial credits. They introduce a similarity ranking method instead of a traditional Word Sense Disambiguation (WSD) approach; which evaluates how well the model ranks different uses of a word based on context, particularly for rare senses, ensuring that occurrences with the same meaning appear closer together in the ranking.

1.2 Methodology

To implement their ranking approach, authors of [1] utilizes two sense-annotated corpora: a larger reference corpus (D) containing multiple annotated examples of the same word in different contexts, and a smaller query corpus (Q) where specific occurrences of the word serve as queries. A sentence from Q containing a target word serves as a query to retrieve other sentences in the corpus D where the same word appears, even if it carries different meanings depending on the context.

To evaluate a given model, the contextual embedding of each target word in Q is extracted. With the same model, for each sentence in D the embedding of the target word is then extracted. Finally, using the cosine similarity, the examples in D are ranked based on their similarity to the query.

Performance is evaluated using Mean Average Precision (MAP) on the top 50 results, and compared to a random baseline and an oracle ranking. A filtering step removes test instances whose label (lemma + sense) appears fewer than 5 times in the training set, ensuring that all test labels are at least minimally represented in the training data. To improve performance, an additional experiment tests whether light fine-tuning on an external dataset helps capture rare word senses more effectively.

To analyze the impact of word frequency and sense ambiguity on model performance, the paper categorizes words into four bins based on frequency ℓ and rarity r. The frequency ℓ of a lemma in the reference corpus D represents how often a word appears in the dataset. Rarity r measures how often a specific label is assigned to a single lemma in the reference corpus D. A high rarity ($r \approx 1$) means that a lemma overwhelmingly appears with the same sense, while a low rarity ($r \approx 0$) means that the sense is shared across multiple lemmas, making it less specific and therefore more ambiguous.

1.3 Reproducibility of the experiments

In this section, we replicate the experiments conducted in [1]. Full results are presented in Appendix A. Overall, our experiments converge with the authors' conclusions. Several pre-trained models — BERT, RoBERTa, DistilBERT, ALBERT, XLNet, GPT-2 and DistilROBERTa — are tested on two English sense-annotated corpora: OntoNotes 5.0 (covering nouns and verbs) and PDEP (Pattern Dictionary of English Prepositions). The results show that all models outperform random ranking, but performance varies significantly, especially for rare senses. BERT achieves the best overall performance, while RoBERTa, despite excelling in other benchmarks (e.g., GLUE [3]), struggles with rare sense representation. This may stem from BERT's training approach, which includes next sentence prediction (NSP) alongside masked language modeling (MLM). In contrast, RoBERTa removes NSP and uses dynamic masking, which might weaken rare sense representation. Studies such as [4] have also shown BERT's superior performance in a WSD task.

2 Extensions to the original paper

2.1 Analysis of the performance of the models for different layers

2.1.1 Motivation

In [1], authors decided to use the embeddings from the **last layer** of each model, based on the assumption that these embeddings capture more task-relevant information for downstream applications. However, numerous studies ([5]) have shown that middle layers tend to better represent syntactic structures, while lower layers are more informative about word order. In this section, we propose a new experiment to assess the quality of the embedding produced by each layer of a model (with respect to WSD).

2.1.2 Methodology

Building on the code available at `https://github.com/lgessler/bert-has-uncommon-sense`, we modified it so that, for each model, we can run the experiments of section 1.3 across all individual layers to assess layer-wise performance. Since the fine-tuning only affect the last layers of the models, we have decided to only consider base models. We are therefore running 82 experiments: BERT, RoBERTa, ALBERT, XLNet, and GPT-2 have 12 layers while DistilBERT and DistilROBERTa have 6 layers.

2.1.3 Results

Full results are presented in Appendix B. The later layers generally yield better performance across all frequency and rarity conditions. However, there are two notable exceptions: first, **GPT-2**, for which the early layers tend to perform better, which aligns with its architecture designed for text generation tasks. As a decoder-only model, GPT-2's later layers become highly specialized for next-token prediction, making them less effective for tasks requiring general linguistic understanding. Secondly, **XLNet** shows better performance in the earlier layers due to its particular training objective that incorporates permutations of the input sequence. This enables it to encode linguistic knowledge more effectively in earlier layers [6].

2.2 Robustness to a change in metric

2.2.1 Motivation

While [1] use cosine similarity to rank contextualized word embeddings, it remains an unanswered question how sensitive their framework is to the choice of similarity metric. Cosine similarity is often used in embedding-based NLP tasks because it focuses on the direction of vectors while ignoring their amplitude. However, when working with models that do not explicitly normalize their output embeddings, the norms of the embedding vectors may encode meaningful information. To assess the robustness of the proposed ranking framework, we investigate the effect of replacing cosine similarity with Euclidean distance. This allows us to test whether the model's ability to disambiguite word senses depends critically on angular distance, or whether meaningful desambiguation persists under alternative geometric assumptions. Such an analysis also helps clarify whether certain word sense distinctions are encoded more in direction or in magnitude.

2.2.2 Methodology

Building on the code available at `https://github.com/lgessler/bert-has-uncommon-sense`, we modified it so that, for each model, we can run the experiments of section 1.3 but using the euclidean distance as a similarity measure instead of the cosine similarity. Due to computation limits, we have decided to only run the experiments for the PDEP dataset. We are therefore running 35 experiments, testing 7 models on the 5 levels of fine-tuning.

2.3 Results

Full results are presented in Appendix C. They highlight only minor differences between cosine similarity and euclidean distance. Surprisingly, Euclidean distance gives results that are comparable to, and in some cases better than cosine similarity; particularly in the fine-tuning scenario with DistilRoBERTa, where it performs better on low-frequency and rare word senses.

2.4 Experiments on French model and corpora

2.4.1 Motivation

The paper focuses on English models and benchmarks, which are well-studied due to abundant data, but little is known about French model performance. However, French presents additional linguistic challenges, making English models unsuitable due to segmentation issues and limited exposure to French grammar and vocabulary. To evaluate WSD performance for French, we use CamemBERT-base embeddings from layer 11, following the original paper. The model is tested on two French corpora, FLUE and EuroSense, requiring modifications to the existing code for integration.

2.4.2 Datasets and methodology overview

FLUE dataset

FLUE [7],[8] is a French NLP benchmark, inspired by GLUE for English, and based on FrenchSemEval (FSE) for Verb Sense Disambiguation (VSD). Since the training set contains multiple word types while the test set includes only verbs, we filter the training set to retain only verbs to reduce computational costs. Unlike PDEP and OntoNotes, FLUE has much lower frequency counts, with its most frequent label appearing only 24 times (compared to 962 in PDEP). The original paper's fixed frequency bins ($\ell = [5, 500[, [500, 1e9[$) do not suit FLUE, as only 4.5% of its data exceed five occurrences. To adapt, we redefine frequency bins based on quantiles and the median ($\ell = [1, 2[, [2, 24[$) , remove the filter that excludes rare instances, and ignore rarity, which becomes meaningless due to FLUE's low word frequencies. The code is also modified to handle empty bins. After adapting the code for FLUE, we present the results in the following section. A limitation of basing the analysis on the frequency ℓ found in the dataset is that it may not reflect real-world usage. A word might be rare in the dataset but common in actual language, as is the case with FLUE, which questions the validity of frequency-based categorization.

Eurosense dataset

Since FLUE may not fully capture the diversity of the French language due to the dominance of unique words, we turn to EuroSense [9], a multilingual WSD dataset based on Europarl, covering 21 languages with 123 million sense annotations for 155,000+ concepts, using BabelNet as its annotation source. To ensure high-quality annotations, we use the High Precision version, which filters out unreliable annotations. As our focus is on French, the dataset is filtered to retain only French annotations. However, unexpected issues arose, revealing errors in the dataset such as mislabeled languages, empty annotations, and punctuation-only entries, requiring additional data cleaning to ensure reliability. The French dataset is then divided into four subsets (verbs, nouns, adverbs, and adjectives) to compare CamemBERT-base's performance across word types, while a combined dataset is retained for reference. Given the large dataset size and our limited computational resources, stratified subsampling based on quartiles is applied to obtain smaller datasets with a balanced representation of frequent and rare words. We selected 80,000 annotations for each subsample. As the training and test sets were not provided, each subsampled dataset is split into training (80%) and test (20%) sets, also ensuring that both frequent and rare labels are represented in the test set. Finally, since EuroSense contains multi-word annotations (e.g., "avoir été"), which the original paper's code did not support, the code is adapted to average the embeddings of each word in the annotation. An option to exclude multi-word annotations is also introduced to enable comparisons between models trained with and without them. After adapting the code for EuroSense, predictions are generated and presented in the results section.

2.4.3 Results

FLUE is not directly comparable to other datasets due to different frequency bins. However, frequent words (17% precision) are better disambiguated than rare ones (5%). Overall performance remains low, likely because FLUE mostly contains rare words, making it harder for the model to find similar words in the corpus. For **EuroSense**, nouns have the highest precision, while verbs perform the worst across all bins. Adjectives and adverbs have lower precision than nouns, but their ranking varied across runs. French nouns could be easier to disambiguate due to gender agreement and syntactic structure, while verbs are harder because of complex conjugation and polysemy. Adjectives and adverbs could remain challenging due to flexible word order and morphological variations. Furthermore, as in OntoNotes and PDEP, rarity was the primary factor affecting performance. High-rarity words had better average precision (74% for high-frequency words, 64% for low-frequency words), while low-rarity words performed significantly worse (17% and 12%, respectively). While higher frequency improved performance, its effect was less significant than rarity. This could be explained by the model's difficulty with low-rarity labels, where the weak correlation between the lemma and its sense increased ambiguity.

3 Conclusion

In this report, we have successfully reproduced the experiments from [1], confirming that BERT-like encoder models can effectively disentangle rare word senses. Their large architectures and extensive training data contribute to strong performance on PDEP and Ontonotes. However, these models are not equal in performance, and, despite excellent results on other benchmark (e.g. GLUE), ROBERTa achieves here poor results compared to other models. Building on these findings, we explored three new directions. First, our analysis of embedding layers shows that while later layers usually provide better performance for encoder models, the behavior of decoder models like GPT-2 and models with unique training objectives like XLNet can differ significantly. Furthermore, we have shown that, surprisingly, the choice of the euclidean metric instead of the cosine similarity does not hamper the performances. Finally, we have concluded with a very promising extension of G&S' work, extending it to French datasets and models.

Contributions

For this project, we divided the work as follows:

- **Abdellah**: He first ran the reproduction of the paper's experiments on the whole PDEP and part of the OntoNotes dataset. He then ran the experiments on the impact of the layer used to extract the embeddings on the PDEP dataset. Additionally, inspired by the paper [10], he worked on a deep analysis of the PDEP dataset. The PDEP dataset was actually built by merging three different datasets: OEC, FN, and CPA. While OEC is of high quality (built by a consortium of experts), CPA was built using data synthesis. Therefore, CPA's quality is not comparable to OEC's quality. And actually, when conducting the analysis of the PDEP dataset (three scripts were added to the original repository here `https://github.com/LilaR66/project-word-sense-disambiguation` to conduct this analysis; they are present in the "pdep_in_depth" folder), two interesting points can be raised: first, in G&S's paper, lots of training sentences from CPA and FN that were badly tagged were still kept; second, sentences that are in FN (which represent 60 percent of the PDEP dataset used by G&S) tend to always be "near" (in terms of embedding) other sentences in FN (compared to sentences in OEC and CPA), despite not necessarily having the same target word and the same target sense (FN sentences are written in a particular style?). The results of these analyses were not added to the report because Abdellah thought that he was still lacking perspective on these results, and not enough analysis of them had been conducted. Finally, Abdellah conducted the experiments and analysed the results of the section about the Euclidean distance.

- **David:** He conducted an independent **reproduction of the paper's results** on the PDEP and OntoNotes datasets, verifying the consistency of the findings. In doing so, he systematically compared the reproduced results with those reported in the original paper and those previously obtained by Abdellah, which led him to the identification of an error in the original paper's appendix—specifically, certain tables were erroneously replicated from a single case. As his main task, David performed an extensive analysis of **model performance across different layers**, gathering and evaluating empirical evidence to understand variations in performance. Based on these observations, he investigated and explained the reasons behind the observed discrepancies. To enhance the interpretability of these findings, he generated plots illustrating the mean average precision across different conditions, revealing recurring patterns in performance differences.

- **Lila**: She began by reproducing experiments on the CLRES dataset, comparing her results with those obtained by Abdellah and David. She then attempted to adapt the method from the original paper to more recent models such as DeepSeek and Mistral. However, this proved to be a major challenge due to the fact that the entire pipeline (from data preprocessing to model evaluation) relies heavily on the latest version of the AllenNLP library, which is not compatible with newer tokenizers like LlamaTokenizer. Lila explored and tested numerous workarounds to overcome this incompatibility — including substituting tokenizers, manually adjusting tokenization outputs, and attempting to decouple specific components of the pipeline from AllenNLP. Despite these efforts, no solution proved effective without requiring a deep and extensive refactoring of the entire codebase. Given the time constraints she ultimately decided to abandon this line of investigation and explore a more feasible alternative. Following this, Lila focused on extending the project to work with French data. She chose a model supported by AllenNLP — CamemBERT-base — and applied it to two new French WSD datasets. This extension required significant adaptation of the existing codebase. Specifically, she made modifications to the four main scripts and developed 15 new scripts to handle various aspects of the new pipeline: cleaning and preprocessing the French datasets, formatting them to match AllenNLP's expected input structure, integrating them into the pipeline, and addressing several technical challenges such as the handling of multi-word expressions in embeddings and the management of empty bins in the evaluation phase. She then ran 24 experiments on the two French corpora and analyzed the results in detail. All of the work — including code changes, new scripts, and results — is fully documented in the GitHub repository: `https://github.com/LilaR66/project-word-sense-disambiguation`. The README files in the "bert-has-uncommon-sense" folder includes clear instructions on how to set up the environment and reproduce the experiments, along with a detailed summary of all changes made.

References

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Appendix

A Reproduction of the paper's experiments

Model	$\ell < 500$		$\ell > 500$	
	$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
random-baseline	11.54	62.39	9.56	76.09
oracle	82.02	93.89	100.0	100.0
bert-base-cased	41.60	81.89	48.48	88.53
distilbert-base-cased	39.80	81.32	48.17	88.50
roberta-base	32.87	78.39	45.16	88.37
distilroberta-base	29.33	76.48	43.69	86.86
albert-base-v2	40.44	81.81	51.58	89.56
xlnet-base-cased	28.71	75.07	36.16	84.29
gpt2	18.33	69.56	33.44	82.70

Model	$\ell < 500$		$\ell > 500$	
	$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
random-baseline	11.54	62.39	9.56	76.09
oracle	82.02	93.89	100.0	100.0
bert-base-cased	43.57	82.54	51.26	90.10
distilbert-base-cased	41.96	82.03	51.28	89.38
roberta-base	37.39	80.53	49.51	89.64
distilroberta-base	33.39	78.94	48.16	88.54
albert-base-v2	40.44	81.81	51.58	89.56
xlnet-base-cased	38.45	80.03	53.06	87.83
gpt2	18.53	69.56	33.44	82.70

Table 1: **(a)** Mean Average Precision (MAP) for Ontonotes 5.0, no fine-tuning, with cosine similarity (reproduction of the paper). **(b)** Best MAP across all fine-tuning trials for each model on Ontonotes 5.0, with cosine similarity (reproduction of the paper).

Model	$\ell < 500$		$\ell > 500$	
	$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
random-baseline	11.80	56.23	18.27	50.03
oracle	96.41	100.0	100.0	100.0
bert-base-cased	59.59	83.54	66.34	89.39
distilbert-base-cased	58.06	83.14	65.09	88.09
roberta-base	39.84	76.79	47.65	80.00
distilroberta-base	32.42	72.22	42.29	70.81
albert-base-v2	56.53	82.22	66.64	88.44
xlnet-base-cased	35.76	74.08	37.68	75.16
gpt2	21.76	63.37	33.33	61.01

Model	$\ell < 500$		$\ell > 500$	
	$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
random-baseline	11.80	56.23	18.27	50.03
oracle	96.41	100.0	100.0	100.0
bert-base-cased	60.57	83.87	68.26	90.00
distilbert-base-cased	57.80	83.56	67.09	88.09
roberta-base	49.84	80.25	57.15	85.33
distilroberta-base	40.78	76.77	50.97	77.94
albert-base-v2	56.53	82.22	66.82	88.44
xlnet-base-cased	48.28	80.88	51.46	83.95
gpt2	21.76	63.59	34.98	61.01

Table 2: **(a)** Mean Average Precision (MAP) for PDEP, no fine-tuning, with cosine similarity (reproduction of the paper). **(b)** Best MAP across all fine-tuning trials for each model on PDEP, with cosine similarity (reproduction of the paper).

B Results of the experiments on different layers

B.1 Numerical results

Model	Layer	MAP	Mean recall	Mean F1-score
random baseline	-	11.46	3.61	5.16
oracle	-	82.02	23.92	34.59
bert-base-cased	0	20.62	6.08	8.76
bert-base-cased	1	30.63	8.63	12.51
bert-base-cased	2	34.93	9.91	14.36
bert-base-cased	3	37.02	10.55	15.28
bert-base-cased	4	38.93	11.10	16.08
bert-base-cased	5	40.32	11.50	16.66
bert-base-cased	6	42.13	12.03	17.43
bert-base-cased	7	42.11	12.02	17.41
bert-base-cased	8	42.14	12.00	17.39
bert-base-cased	9	42.24	12.00	17.40
bert-base-cased	10	41.65	11.81	17.12
bert-base-cased	11	41.60	11.79	17.09
distilbert-base-cased	0	26.14	7.47	10.81
distilbert-base-cased	1	33.01	9.44	13.66
distilbert-base-cased	2	36.15	10.36	14.99
distilbert-base-cased	3	37.95	10.83	15.68
distilbert-base-cased	4	39.69	11.27	16.33
distilbert-base-cased	5	39.80	11.26	16.33
roberta-base	0	19.59	5.79	8.34
roberta-base	1	22.78	6.64	9.59
roberta-base	2	24.03	6.97	10.06
roberta-base	3	25.01	7.24	10.45
roberta-base	4	26.50	7.63	11.03
roberta-base	5	27.70	7.96	11.51
roberta-base	6	29.29	8.40	12.15
roberta-base	7	29.89	8.56	12.39
roberta-base	8	32.06	9.17	13.27
roberta-base	9	34.09	9.70	14.05
roberta-base	10	34.37	9.76	14.14
roberta-base	11	32.87	9.28	13.45
distilroberta-base	0	20.45	6.01	8.66
distilroberta-base	1	23.42	6.80	9.82
distilroberta-base	2	24.74	7.12	10.30
distilroberta-base	3	26.54	7.59	10.99
distilroberta-base	4	29.74	8.44	12.23
distilroberta-base	5	29.33	8.30	12.02
albert-base-v2	0	25.27	7.28	10.52
albert-base-v2	1	30.40	8.72	12.62
albert-base-v2	2	33.93	9.74	14.09
albert-base-v2	3	36.06	10.36	14.98
albert-base-v2	4	37.32	10.71	15.51
albert-base-v2	5	37.99	10.91	15.79
albert-base-v2	6	38.27	10.98	15.89
albert-base-v2	7	38.37	10.99	15.91

albert-base-v2	8	38.25	10.94	15.84
albert-base-v2	9	38.15	10.91	15.79
albert-base-v2	10	38.82	11.09	16.07
albert-base-v2	11	40.44	11.50	16.66
xlnet-base-cased	0	28.33	7.98	11.57
xlnet-base-cased	1	35.52	10.08	14.61
xlnet-base-cased	2	38.75	11.04	16.00
xlnet-base-cased	3	40.57	11.60	16.79
xlnet-base-cased	4	39.88	11.37	16.47
xlnet-base-cased	5	37.47	10.65	15.44
xlnet-base-cased	6	36.75	10.44	15.12
xlnet-base-cased	7	34.87	9.86	14.30
xlnet-base-cased	8	34.77	9.83	14.26
xlnet-base-cased	9	35.97	10.16	14.73
xlnet-base-cased	10	33.22	9.24	13.43
xlnet-base-cased	11	28.72	7.93	11.53
gpt2	0	18.13	5.36	7.72
gpt2	1	19.95	5.82	8.40
gpt2	2	21.98	6.38	9.21
gpt2	3	23.01	6.68	9.65
gpt2	4	23.41	6.80	9.81
gpt2	5	23.63	6.86	9.91
gpt2	6	23.84	6.92	9.99
gpt2	7	23.81	6.90	9.97
gpt2	8	23.91	6.93	10.01
gpt2	9	23.85	6.92	9.99
gpt2	10	23.33	6.77	9.77
gpt2	11	18.34	5.37	7.74

Table 3: MAP, Mean recall and Mean F1-score for low frequency and low rarity conditions for OntoNotes 5.0.

Model	**Layer**	**MAP**	**Mean recall**	**Mean F1-score**
bert-base-cased	0	20.76	3.10	5.16
bert-base-cased	1	36.11	5.37	8.96
bert-base-cased	2	40.04	6.00	9.99
bert-base-cased	3	44.15	6.63	11.04
bert-base-cased	4	48.35	7.25	12.08
bert-base-cased	5	56.29	8.49	14.14
bert-base-cased	6	58.61	8.87	14.76
bert-base-cased	7	60.03	9.09	15.14
bert-base-cased	8	60.27	9.13	15.20
bert-base-cased	9	60.06	9.09	15.14
bert-base-cased	10	59.58	9.00	14.99
bert-base-cased	11	59.59	8.99	14.97
distilbert-base-cased	0	27.70	4.13	6.88
distilbert-base-cased	1	36.90	5.51	9.18
distilbert-base-cased	2	46.78	7.02	11.70
distilbert-base-cased	3	56.07	8.45	14.07
distilbert-base-cased	4	57.80	8.72	14.52
distilbert-base-cased	5	58.06	8.75	14.57

roberta-base	0	17.74	2.64	4.40
roberta-base	1	20.10	3.00	5.00
roberta-base	2	22.76	3.40	5.66
roberta-base	3	24.63	3.68	6.12
roberta-base	4	27.66	4.14	6.90
roberta-base	5	30.93	4.64	7.73
roberta-base	6	34.63	5.19	8.64
roberta-base	7	38.44	5.74	9.57
roberta-base	8	42.78	6.41	10.68
roberta-base	9	43.54	6.51	10.85
roberta-base	10	41.29	6.14	10.24
roberta-base	11	39.84	5.90	9.83
distilroberta-base	0	18.58	2.76	4.59
distilroberta-base	1	21.55	3.21	5.34
distilroberta-base	2	24.34	3.63	6.05
distilroberta-base	3	30.83	4.61	7.68
distilroberta-base	4	32.76	4.88	8.12
distilroberta-base	5	32.42	4.80	8.00
albert-base-v2	0	30.13	4.42	7.37
albert-base-v2	1	39.86	5.96	9.92
albert-base-v2	2	45.44	6.84	11.38
albert-base-v2	3	48.75	7.37	12.26
albert-base-v2	4	50.42	7.64	12.72
albert-base-v2	5	51.37	7.79	12.96
albert-base-v2	6	51.81	7.85	13.07
albert-base-v2	7	52.04	7.90	13.14
albert-base-v2	8	52.18	7.92	13.17
albert-base-v2	9	52.56	7.97	13.27
albert-base-v2	10	53.54	8.12	13.52
albert-base-v2	11	56.53	8.55	14.24
xlnet-base-cased	0	35.65	5.26	8.77
xlnet-base-cased	1	48.03	7.10	11.84
xlnet-base-cased	2	52.76	7.88	13.13
xlnet-base-cased	3	54.69	8.21	13.67
xlnet-base-cased	4	53.65	8.05	13.41
xlnet-base-cased	5	51.71	7.73	12.88
xlnet-base-cased	6	50.58	7.55	12.59
xlnet-base-cased	7	48.40	7.20	12.00
xlnet-base-cased	8	47.25	7.04	11.73
xlnet-base-cased	9	48.36	7.19	11.99
xlnet-base-cased	10	42.67	6.21	10.37
xlnet-base-cased	11	35.76	5.12	8.55
gpt2	0	18.44	2.73	4.54
gpt2	1	20.97	3.08	5.14
gpt2	2	26.86	3.86	6.45
gpt2	3	30.89	4.45	7.43
gpt2	4	32.58	4.73	7.89
gpt2	5	33.76	4.92	8.21
gpt2	6	34.44	5.05	8.42
gpt2	7	34.50	5.07	8.45
gpt2	8	34.12	5.02	8.37

gpt2	9	33.30	4.91	8.19
gpt2	10	31.81	4.69	7.82
gpt2	11	21.76	3.24	5.40

Table 4: MAP, Mean recall and Mean F1-score for low frequency and low rarity conditions for PDEP.

Model	**Layer**	**Mean precision**	**Mean recall**	**Mean F1-score**
random baseline	-	62.56	10.56	17.34
oracle	-	93.89	15.62	25.67
bert-base-cased	0	71.24	11.88	19.52
bert-base-cased	1	76.85	12.73	20.92
bert-base-cased	2	78.68	13.04	21.43
bert-base-cased	3	80.18	13.31	21.87
bert-base-cased	4	81.08	13.46	22.13
bert-base-cased	5	81.78	13.58	22.32
bert-base-cased	6	82.32	13.67	22.47
bert-base-cased	7	82.29	13.66	22.45
bert-base-cased	8	82.28	13.65	22.44
bert-base-cased	9	82.23	13.63	22.41
bert-base-cased	10	82.05	13.60	22.35
bert-base-cased	11	81.89	13.57	22.30
distilbert-base-cased	0	75.44	12.52	20.57
distilbert-base-cased	1	78.16	12.98	21.33
distilbert-base-cased	2	79.83	13.26	21.79
distilbert-base-cased	3	80.65	13.39	22.00
distilbert-base-cased	4	81.35	13.48	22.17
distilbert-base-cased	5	81.32	13.47	22.14
roberta-base	0	70.08	11.71	19.23
roberta-base	1	71.79	11.97	19.66
roberta-base	2	72.82	12.12	19.92
roberta-base	3	73.92	12.29	20.20
roberta-base	4	74.97	12.45	20.47
roberta-base	5	75.65	12.57	20.65
roberta-base	6	76.59	12.71	20.90
roberta-base	7	77.08	12.79	21.02
roberta-base	8	78.15	12.97	21.31
roberta-base	9	79.25	13.14	21.60
roberta-base	10	79.29	13.14	21.60
roberta-base	11	78.39	12.97	21.33
distilroberta-base	0	70.61	11.79	19.36
distilroberta-base	1	72.51	12.07	19.83
distilroberta-base	2	73.55	12.22	20.08
distilroberta-base	3	74.87	12.43	20.42
distilroberta-base	4	76.67	12.71	20.89
distilroberta-base	5	76.48	12.67	20.83
albert-base-v2	0	74.43	12.37	20.33
albert-base-v2	1	77.47	12.87	21.14
albert-base-v2	2	79.21	13.16	21.63
albert-base-v2	3	80.12	13.31	21.88
albert-base-v2	4	80.65	13.40	22.02

albert-base-v2	5	80.95	13.45	22.10
albert-base-v2	6	81.04	13.46	22.13
albert-base-v2	7	81.05	13.46	22.12
albert-base-v2	8	80.94	13.44	22.09
albert-base-v2	9	80.83	13.42	22.06
albert-base-v2	10	81.15	13.48	22.16
albert-base-v2	11	81.81	13.58	22.32
xlnet-base-cased	0	76.12	12.60	20.72
xlnet-base-cased	1	79.16	13.13	21.58
xlnet-base-cased	2	80.74	13.40	22.03
xlnet-base-cased	3	81.42	13.51	22.21
xlnet-base-cased	4	81.19	13.47	22.14
xlnet-base-cased	5	80.52	13.35	21.94
xlnet-base-cased	6	80.11	13.28	21.83
xlnet-base-cased	7	79.42	13.15	21.62
xlnet-base-cased	8	79.42	13.15	21.62
xlnet-base-cased	9	79.72	13.20	21.70
xlnet-base-cased	10	77.49	12.79	21.02
xlnet-base-cased	11	75.07	12.39	20.37
gpt2	0	68.60	11.45	18.80
gpt2	1	70.05	11.66	19.16
gpt2	2	71.88	11.95	19.63
gpt2	3	72.89	12.11	19.91
gpt2	4	73.21	12.16	19.99
gpt2	5	73.49	12.21	20.07
gpt2	6	73.68	12.25	20.12
gpt2	7	73.73	12.26	20.15
gpt2	8	73.84	12.28	20.18
gpt2	9	73.80	12.27	20.16
gpt2	10	73.51	12.22	20.08
gpt2	11	69.56	11.56	19.00

Table 5: MAP, Mean recall and Mean F1-score for low frequency and high rarity conditions for OntoNotes 5.0.

Model	**Layer**	**Mean precision**	**Mean recall**	**Mean F1-score**
bert-base-cased	0	65.44	6.95	12.21
bert-base-cased	1	75.65	8.01	14.07
bert-base-cased	2	77.82	8.26	14.50
bert-base-cased	3	79.53	8.44	14.83
bert-base-cased	4	81.13	8.61	15.13
bert-base-cased	5	82.68	8.79	15.43
bert-base-cased	6	83.51	8.88	15.60
bert-base-cased	7	83.87	8.92	15.67
bert-base-cased	8	84.10	8.95	15.72
bert-base-cased	9	84.22	8.96	15.73
bert-base-cased	10	84.00	8.92	15.67
bert-base-cased	11	83.54	8.86	15.56
distilbert-base-cased	0	70.41	7.47	13.12
distilbert-base-cased	1	75.55	8.00	14.05
distilbert-base-cased	2	79.66	8.46	14.85

distilbert-base-cased	3	82.13	8.73	15.34
distilbert-base-cased	4	83.41	8.86	15.57
distilbert-base-cased	5	83.15	8.83	15.50
roberta-base	0	62.33	6.59	11.58
roberta-base	1	63.53	6.72	11.81
roberta-base	2	65.44	6.92	12.15
roberta-base	3	66.83	7.09	12.45
roberta-base	4	69.29	7.35	12.91
roberta-base	5	71.53	7.58	13.32
roberta-base	6	73.48	7.79	13.68
roberta-base	7	74.91	7.94	13.95
roberta-base	8	77.33	8.20	14.40
roberta-base	9	78.26	8.30	14.57
roberta-base	10	77.28	8.18	14.37
roberta-base	11	76.79	8.12	14.27
distilroberta-base	0	63.11	6.67	11.71
distilroberta-base	1	65.64	6.94	12.20
distilroberta-base	2	67.81	7.18	12.62
distilroberta-base	3	70.77	7.50	13.17
distilroberta-base	4	72.51	7.67	13.48
distilroberta-base	5	72.22	7.63	13.40
albert-base-v2	0	70.96	7.49	13.16
albert-base-v2	1	75.62	8.01	14.07
albert-base-v2	2	77.95	8.27	14.52
albert-base-v2	3	79.20	8.41	14.77
albert-base-v2	4	79.91	8.50	14.92
albert-base-v2	5	80.42	8.55	15.02
albert-base-v2	6	80.68	8.58	15.07
albert-base-v2	7	80.93	8.61	15.11
albert-base-v2	8	81.04	8.62	15.13
albert-base-v2	9	81.10	8.62	15.13
albert-base-v2	10	81.27	8.64	15.17
albert-base-v2	11	82.22	8.73	15.34
xlnet-base-cased	0	73.89	7.83	13.76
xlnet-base-cased	1	79.28	8.41	14.77
xlnet-base-cased	2	81.20	8.62	15.15
xlnet-base-cased	3	81.92	8.70	15.29
xlnet-base-cased	4	81.74	8.67	15.24
xlnet-base-cased	5	81.24	8.62	15.14
xlnet-base-cased	6	80.87	8.58	15.06
xlnet-base-cased	7	80.10	8.50	14.92
xlnet-base-cased	8	79.60	8.44	14.83
xlnet-base-cased	9	80.27	8.51	14.95
xlnet-base-cased	10	77.74	8.20	14.41
xlnet-base-cased	11	74.08	7.80	13.70
gpt2	0	60.92	6.48	11.37
gpt2	1	63.07	6.69	11.75
gpt2	2	65.95	6.97	12.24
gpt2	3	67.46	7.13	12.53
gpt2	4	68.11	7.21	12.67
gpt2	5	68.77	7.28	12.79

gpt2	6	69.52	7.36	12.93
gpt2	7	69.77	7.39	12.98
gpt2	8	69.71	7.38	12.97
gpt2	9	69.31	7.35	12.92
gpt2	10	68.90	7.31	12.84
gpt2	11	63.37	6.71	11.79
gpt2	10	73.51	12.22	20.08
gpt2	11	69.56	11.56	19.00

Table 6: MAP, Mean recall and Mean F1-score for low frequency and high rarity conditions for PDEP.

Model	**Layer**	**Mean precision**	**Mean recall**	**Mean F1-score**
random baseline	-	9.37	0.13	0.26
oracle	-	100.00	1.37	2.70
bert-base-cased	0	23.34	0.30	0.58
bert-base-cased	1	43.55	0.58	1.14
bert-base-cased	2	48.84	0.66	1.29
bert-base-cased	3	48.98	0.66	1.29
bert-base-cased	4	50.79	0.68	1.33
bert-base-cased	5	52.22	0.70	1.38
bert-base-cased	6	55.60	0.75	1.47
bert-base-cased	7	55.32	0.74	1.46
bert-base-cased	8	54.29	0.73	1.43
bert-base-cased	9	51.67	0.69	1.35
bert-base-cased	10	49.27	0.65	1.27
bert-base-cased	11	48.48	0.64	1.25
distilbert-base-cased	0	38.11	0.50	0.98
distilbert-base-cased	1	46.62	0.62	1.22
distilbert-base-cased	2	49.61	0.66	1.30
distilbert-base-cased	3	52.65	0.71	1.39
distilbert-base-cased	4	50.92	0.67	1.33
distilbert-base-cased	5	48.17	0.63	1.24
roberta-base	0	33.30	0.43	0.85
roberta-base	1	38.80	0.52	1.01
roberta-base	2	40.99	0.55	1.08
roberta-base	3	40.96	0.55	1.08
roberta-base	4	42.12	0.56	1.11
roberta-base	5	43.48	0.58	1.14
roberta-base	6	45.19	0.60	1.18
roberta-base	7	45.47	0.60	1.19
roberta-base	8	47.22	0.63	1.24
roberta-base	9	47.84	0.64	1.25
roberta-base	10	46.95	0.62	1.22
roberta-base	11	45.16	0.59	1.17
distilroberta-base	0	36.48	0.48	0.94
distilroberta-base	1	41.19	0.55	1.09
distilroberta-base	2	41.38	0.55	1.09
distilroberta-base	3	44.17	0.59	1.16
distilroberta-base	4	45.15	0.60	1.18
distilroberta-base	5	43.69	0.58	1.13

albert-base-v2	0	42.60	0.57	1.11
albert-base-v2	1	47.38	0.63	1.24
albert-base-v2	2	49.86	0.67	1.31
albert-base-v2	3	50.55	0.68	1.33
albert-base-v2	4	50.75	0.68	1.33
albert-base-v2	5	50.82	0.68	1.34
albert-base-v2	6	50.44	0.68	1.33
albert-base-v2	7	50.13	0.67	1.32
albert-base-v2	8	49.94	0.67	1.31
albert-base-v2	9	49.64	0.66	1.30
albert-base-v2	10	50.39	0.67	1.32
albert-base-v2	11	51.58	0.69	1.35
xlnet-base-cased	0	42.77	0.56	1.11
xlnet-base-cased	1	49.96	0.67	1.31
xlnet-base-cased	2	51.16	0.68	1.34
xlnet-base-cased	3	53.27	0.72	1.41
xlnet-base-cased	4	53.27	0.72	1.41
xlnet-base-cased	5	52.27	0.70	1.38
xlnet-base-cased	6	51.95	0.70	1.37
xlnet-base-cased	7	49.45	0.66	1.30
xlnet-base-cased	8	48.68	0.65	1.28
xlnet-base-cased	9	48.33	0.64	1.26
xlnet-base-cased	10	44.07	0.58	1.14
xlnet-base-cased	11	36.16	0.47	0.91
gpt2	0	25.21	0.34	0.66
gpt2	1	32.20	0.43	0.85
gpt2	2	34.25	0.46	0.91
gpt2	3	34.75	0.47	0.93
gpt2	4	35.32	0.48	0.94
gpt2	5	35.26	0.48	0.94
gpt2	6	35.31	0.48	0.94
gpt2	7	35.61	0.49	0.95
gpt2	8	35.67	0.49	0.95
gpt2	9	35.66	0.49	0.96
gpt2	10	35.84	0.49	0.96
gpt2	11	33.44	0.45	0.89

Table 7: MAP, Mean recall and Mean F1-score for high frequency and low rarity conditions for OntoNotes 5.0.

Model	Layer	Mean precision	Mean recall	Mean F1-score
bert-base-cased	0	25.48	0.80	1.53
bert-base-cased	1	40.05	1.27	2.44
bert-base-cased	2	45.13	1.44	2.75
bert-base-cased	3	48.70	1.54	2.96
bert-base-cased	4	50.95	1.62	3.10
bert-base-cased	5	62.85	2.01	3.85
bert-base-cased	6	68.70	2.20	4.22
bert-base-cased	7	70.97	2.27	4.36
bert-base-cased	8	70.23	2.25	4.31
bert-base-cased	9	68.28	2.18	4.18

bert-base-cased	10	66.47	2.11	4.05
bert-base-cased	11	66.34	2.11	4.04
distilbert-base-cased	0	33.30	1.06	2.02
distilbert-base-cased	1	44.53	1.41	2.70
distilbert-base-cased	2	52.92	1.68	3.22
distilbert-base-cased	3	67.11	2.14	4.11
distilbert-base-cased	4	66.12	2.10	4.03
distilbert-base-cased	5	65.09	2.06	3.96
roberta-base	0	27.65	0.87	1.67
roberta-base	1	29.86	0.94	1.81
roberta-base	2	31.11	0.98	1.88
roberta-base	3	32.53	1.02	1.95
roberta-base	4	33.82	1.07	2.04
roberta-base	5	36.98	1.17	2.24
roberta-base	6	41.64	1.32	2.53
roberta-base	7	46.91	1.49	2.85
roberta-base	8	53.14	1.69	3.24
roberta-base	9	53.23	1.70	3.26
roberta-base	10	49.82	1.58	3.03
roberta-base	11	47.65	1.51	2.89
distilroberta-base	0	31.24	0.98	1.89
distilroberta-base	1	32.75	1.03	1.98
distilroberta-base	2	32.85	1.04	2.00
distilroberta-base	3	40.88	1.29	2.48
distilroberta-base	4	43.61	1.38	2.65
distilroberta-base	5	42.29	1.34	2.57
albert-base-v2	0	40.14	1.26	2.41
albert-base-v2	1	50.16	1.58	3.04
albert-base-v2	2	55.57	1.76	3.38
albert-base-v2	3	59.12	1.88	3.61
albert-base-v2	4	59.94	1.91	3.66
albert-base-v2	5	60.12	1.92	3.68
albert-base-v2	6	59.96	1.91	3.67
albert-base-v2	7	60.17	1.92	3.69
albert-base-v2	8	60.33	1.93	3.70
albert-base-v2	9	60.44	1.94	3.71
albert-base-v2	10	61.73	1.98	3.79
albert-base-v2	11	66.64	2.13	4.09
xlnet-base-cased	0	46.36	1.46	2.81
xlnet-base-cased	1	60.66	1.92	3.68
xlnet-base-cased	2	65.69	2.08	3.98
xlnet-base-cased	3	69.02	2.20	4.22
xlnet-base-cased	4	66.39	2.12	4.06
xlnet-base-cased	5	64.03	2.04	3.92
xlnet-base-cased	6	61.77	1.96	3.76
xlnet-base-cased	7	60.26	1.91	3.67
xlnet-base-cased	8	57.10	1.81	3.47
xlnet-base-cased	9	55.61	1.75	3.36
xlnet-base-cased	10	47.46	1.48	2.84
xlnet-base-cased	11	37.68	1.15	2.22
gpt2	0	27.00	0.82	1.58

gpt2	1	31.59	0.96	1.85
gpt2	2	42.64	1.29	2.47
gpt2	3	51.34	1.57	3.01
gpt2	4	54.49	1.68	3.21
gpt2	5	56.96	1.76	3.38
gpt2	6	56.99	1.77	3.39
gpt2	7	55.58	1.73	3.32
gpt2	8	54.12	1.68	3.23
gpt2	9	51.59	1.60	3.07
gpt2	10	49.31	1.53	2.93
gpt2	11	33.33	1.04	1.99
gpt2	10	35.84	0.49	0.96
gpt2	11	33.44	0.45	0.89

Table 8: MAP, Mean recall and Mean F1-score for high frequency and low rarity conditions for PDEP.

Model	Layer	Mean precision	Mean recall	Mean F1-score
random baseline	-	76.17	0.26	0.51
oracle	-	100.00	0.34	0.67
bert-base-cased	0	80.11	0.27	0.53
bert-base-cased	1	86.38	0.29	0.58
bert-base-cased	2	87.43	0.29	0.58
bert-base-cased	3	87.81	0.29	0.59
bert-base-cased	4	88.34	0.30	0.59
bert-base-cased	5	88.90	0.30	0.59
bert-base-cased	6	89.65	0.30	0.60
bert-base-cased	7	89.43	0.30	0.60
bert-base-cased	8	89.37	0.30	0.60
bert-base-cased	9	89.00	0.30	0.59
bert-base-cased	10	88.78	0.30	0.59
bert-base-cased	11	88.53	0.30	0.59
distilbert-base-cased	0	84.86	0.28	0.57
distilbert-base-cased	1	86.92	0.29	0.58
distilbert-base-cased	2	88.41	0.30	0.59
distilbert-base-cased	3	88.87	0.30	0.59
distilbert-base-cased	4	88.97	0.30	0.59
distilbert-base-cased	5	88.50	0.30	0.59
roberta-base	0	81.96	0.27	0.55
roberta-base	1	84.24	0.28	0.56
roberta-base	2	85.09	0.29	0.57
roberta-base	3	84.97	0.28	0.57
roberta-base	4	85.72	0.29	0.57
roberta-base	5	86.29	0.29	0.58
roberta-base	6	87.24	0.29	0.58
roberta-base	7	87.72	0.29	0.59
roberta-base	8	88.56	0.30	0.59
roberta-base	9	88.80	0.30	0.59
roberta-base	10	88.59	0.30	0.59
roberta-base	11	88.37	0.30	0.59
distilroberta-base	0	83.32	0.28	0.56

distilroberta-base	1	85.05	0.29	0.57
distilroberta-base	2	85.38	0.29	0.57
distilroberta-base	3	86.23	0.29	0.58
distilroberta-base	4	87.12	0.29	0.58
distilroberta-base	5	86.86	0.29	0.58
albert-base-v2	0	85.43	0.29	0.57
albert-base-v2	1	87.45	0.29	0.58
albert-base-v2	2	88.31	0.30	0.59
albert-base-v2	3	88.82	0.30	0.59
albert-base-v2	4	88.95	0.30	0.59
albert-base-v2	5	88.99	0.30	0.59
albert-base-v2	6	89.06	0.30	0.60
albert-base-v2	7	88.98	0.30	0.59
albert-base-v2	8	88.96	0.30	0.59
albert-base-v2	9	88.94	0.30	0.59
albert-base-v2	10	89.06	0.30	0.60
albert-base-v2	11	89.56	0.30	0.60
xlnet-base-cased	0	85.84	0.29	0.57
xlnet-base-cased	1	87.78	0.29	0.59
xlnet-base-cased	2	88.06	0.30	0.59
xlnet-base-cased	3	88.49	0.30	0.59
xlnet-base-cased	4	88.18	0.30	0.59
xlnet-base-cased	5	88.09	0.30	0.59
xlnet-base-cased	6	87.89	0.29	0.59
xlnet-base-cased	7	88.09	0.30	0.59
xlnet-base-cased	8	87.88	0.29	0.59
xlnet-base-cased	9	88.29	0.30	0.59
xlnet-base-cased	10	87.76	0.29	0.59
xlnet-base-cased	11	84.29	0.28	0.56
gpt2	0	80.12	0.27	0.53
gpt2	1	82.55	0.28	0.55
gpt2	2	83.25	0.28	0.56
gpt2	3	83.69	0.28	0.56
gpt2	4	83.66	0.28	0.56
gpt2	5	83.87	0.28	0.56
gpt2	6	83.88	0.28	0.56
gpt2	7	83.86	0.28	0.56
gpt2	8	83.82	0.28	0.56
gpt2	9	83.80	0.28	0.56
gpt2	10	83.61	0.28	0.56
gpt2	11	82.70	0.28	0.55

Table 9: MAP, Mean recall and Mean F1-score for high frequency and high rarity conditions for OntoNotes 5.0.

Model	Layer	Mean precision	Mean recall	Mean F1-score
bert-base-cased	0	57.18	1.92	3.68
bert-base-cased	1	73.17	2.49	4.77
bert-base-cased	2	76.09	2.60	4.97
bert-base-cased	3	80.35	2.75	5.26
bert-base-cased	4	82.36	2.82	5.39

bert-base-cased	5	88.30	3.03	5.80
bert-base-cased	6	89.47	3.07	5.88
bert-base-cased	7	90.18	3.10	5.93
bert-base-cased	8	90.14	3.10	5.93
bert-base-cased	9	89.89	3.09	5.91
bert-base-cased	10	89.45	3.07	5.88
bert-base-cased	11	89.39	3.07	5.87
distilbert-base-cased	0	63.48	2.16	4.12
distilbert-base-cased	1	72.47	2.47	4.72
distilbert-base-cased	2	80.87	2.76	5.28
distilbert-base-cased	3	88.36	3.04	5.81
distilbert-base-cased	4	88.41	3.04	5.81
distilbert-base-cased	5	88.10	3.02	5.78
roberta-base	0	56.79	1.93	3.69
roberta-base	1	59.43	2.01	3.84
roberta-base	2	61.85	2.10	4.01
roberta-base	3	63.32	2.15	4.12
roberta-base	4	66.28	2.26	4.32
roberta-base	5	69.92	2.39	4.57
roberta-base	6	73.33	2.51	4.80
roberta-base	7	76.75	2.63	5.03
roberta-base	8	80.92	2.77	5.31
roberta-base	9	82.19	2.81	5.38
roberta-base	10	81.21	2.78	5.31
roberta-base	11	80.01	2.73	5.22
distilroberta-base	0	57.93	1.96	3.75
distilroberta-base	1	60.06	2.04	3.90
distilroberta-base	2	61.21	2.09	4.00
distilroberta-base	3	68.13	2.33	4.45
distilroberta-base	4	71.32	2.43	4.65
distilroberta-base	5	70.81	2.42	4.62
albert-base-v2	0	71.37	2.41	4.62
albert-base-v2	1	78.89	2.69	5.15
albert-base-v2	2	82.30	2.82	5.39
albert-base-v2	3	84.55	2.90	5.55
albert-base-v2	4	85.53	2.93	5.61
albert-base-v2	5	86.16	2.96	5.66
albert-base-v2	6	86.41	2.97	5.68
albert-base-v2	7	86.45	2.97	5.68
albert-base-v2	8	86.52	2.97	5.69
albert-base-v2	9	86.71	2.98	5.70
albert-base-v2	10	87.21	3.00	5.74
albert-base-v2	11	88.44	3.04	5.82
xlnet-base-cased	0	71.37	2.43	4.64
xlnet-base-cased	1	82.86	2.83	5.41
xlnet-base-cased	2	86.39	2.96	5.66
xlnet-base-cased	3	87.91	3.01	5.76
xlnet-base-cased	4	87.50	3.00	5.74
xlnet-base-cased	5	87.10	2.98	5.71
xlnet-base-cased	6	86.47	2.96	5.67
xlnet-base-cased	7	85.46	2.93	5.61

xlnet-base-cased	8	85.16	2.92	5.59
xlnet-base-cased	9	85.51	2.93	5.61
xlnet-base-cased	10	82.60	2.82	5.39
xlnet-base-cased	11	75.16	2.52	4.82
gpt2	0	54.24	1.85	3.54
gpt2	1	56.26	1.92	3.67
gpt2	2	61.04	2.07	3.95
gpt2	3	64.59	2.18	4.17
gpt2	4	65.78	2.22	4.24
gpt2	5	67.65	2.29	4.37
gpt2	6	69.34	2.35	4.50
gpt2	7	69.88	2.38	4.55
gpt2	8	69.70	2.38	4.55
gpt2	9	69.68	2.37	4.54
gpt2	10	69.16	2.35	4.51
gpt2	11	61.01	2.08	3.97
gpt2	10	83.61	0.28	0.56
gpt2	11	82.70	0.28	0.55

Table 10: MAP, Mean recall and Mean F1-score for high frequency and high rarity conditions for PDEP.

B.2 Mean average precision (MAP) plots

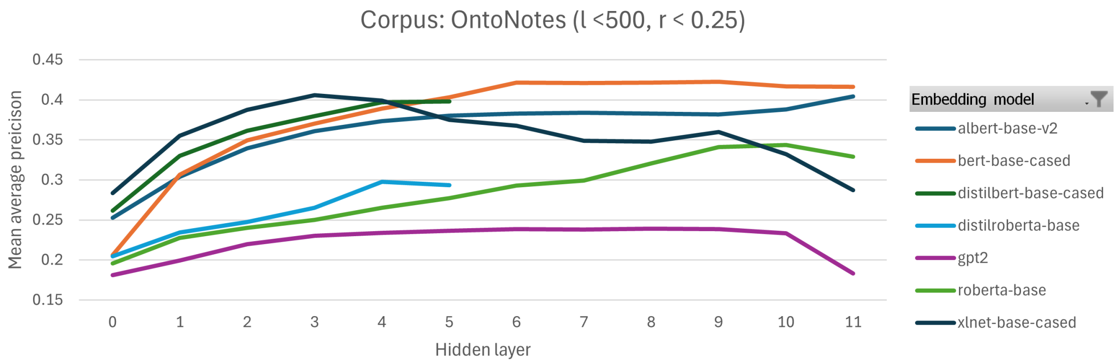

Figure 1: MAP across layers for low frequency and low rarity conditions for OntoNotes 5.0.

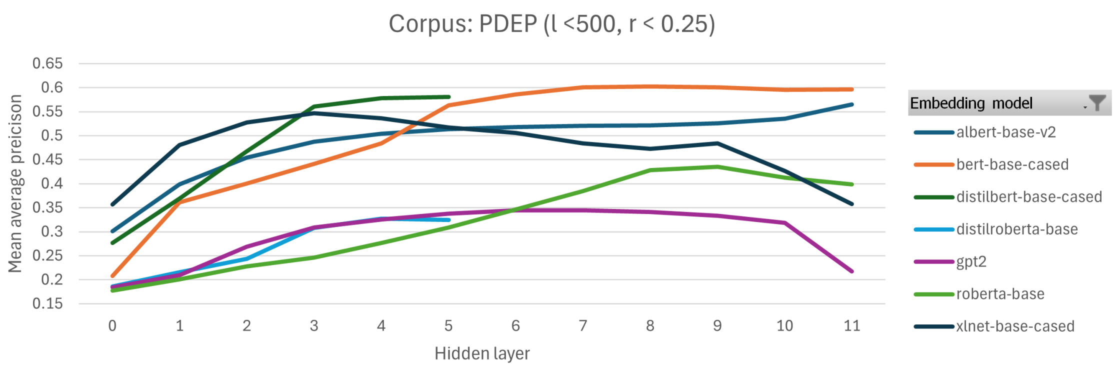

Figure 2: MAP across layers for low frequency and low rarity conditions for PDEP.

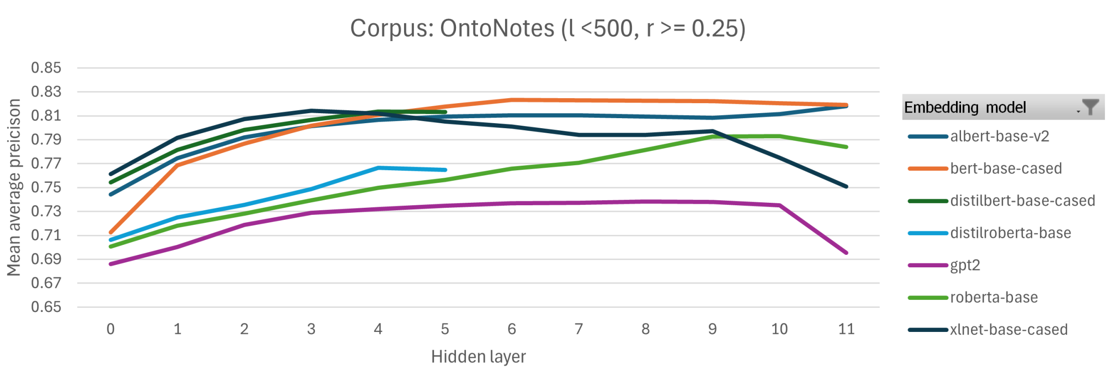

Figure 3: MAP across layers for low frequency and high rarity conditions for OntoNotes 5.0.

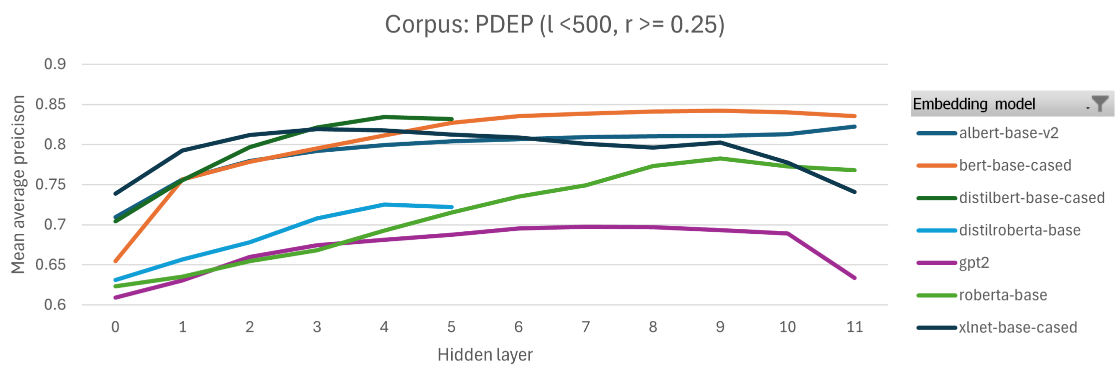

Figure 4: MAP across layers for low frequency and high rarity conditions for PDEP.

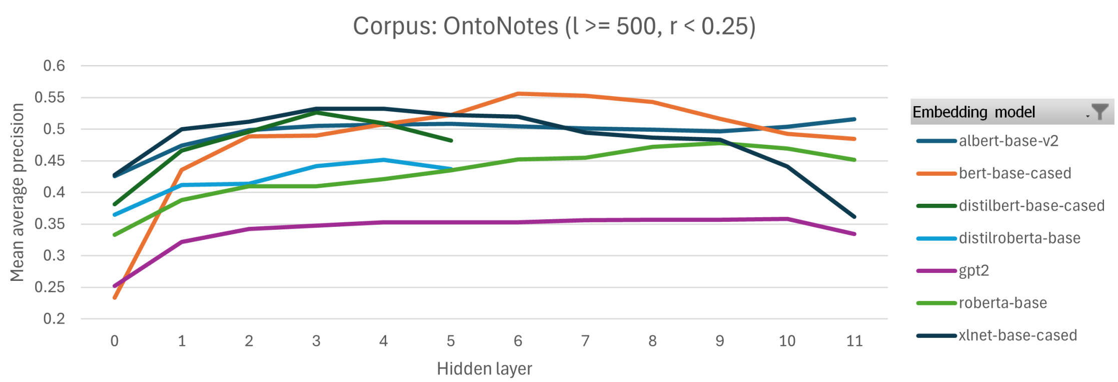

Figure 5: MAP across layers for high frequency and low rarity conditions for OntoNotes 5.0.

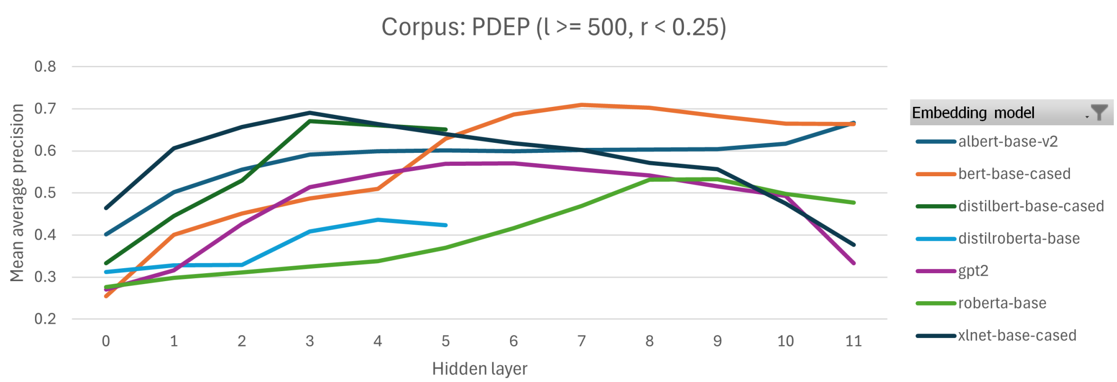

Figure 6: MAP across layers for high frequency and low rarity conditions for PDEP.

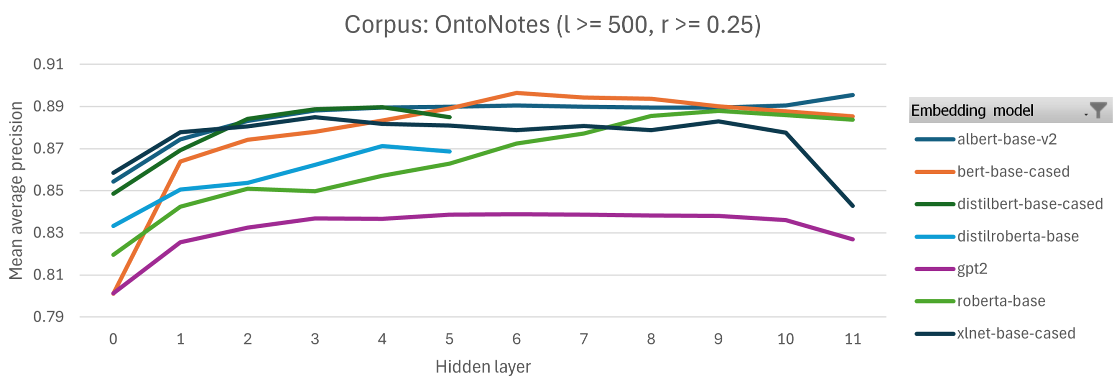

Figure 7: MAP across layers for high frequency and high rarity conditions for OntoNotes 5.0.

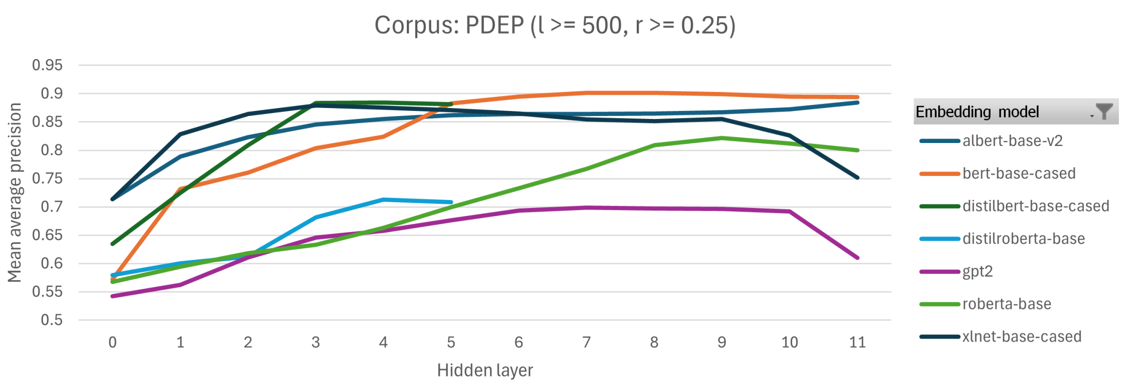

Figure 8: MAP across layers for high frequency and high rarity conditions for PDEP.

C Results of the experiments on the Euclidean distance

Model	$\ell < 500$		$\ell > 500$	
	$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
bert-base-cased	57.95	83.13	65.17	89.28
distilbert-base-cased	56.85	82.85	64.07	88.12
roberta-base	38.78	76.00	46.36	78.41
distilroberta-base	32.11	71.96	41.91	70.13
albert-base-v2	56.26	81.97	66.42	88.40
xlnet-base-cased	29.68	71.50	34.92	69.17
gpt2	19.84	62.01	30.35	59.08

Model	$\ell < 500$		$\ell > 500$	
	$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
bert-base-cased	59.58	83.68	68.01	89.66
distilbert-base-cased	59.49	83.60	66.66	88.39
roberta-base	49.64	79.75	57.51	85.23
distilroberta-base	40.88	77.29	50.96	77.92
albert-base-v2	56.26	81.97	66.42	88.40
xlnet-base-cased	49.06	79.84	57.63	85.90
gpt2	21.26	62.86	35.89	60.20

Table 11: **(a)** Mean Average Precision (MAP) for PDEP, no fine-tuning, with euclidean similarity. **(b)** Best MAP across all fine-tuning trials for each model on PDEP, with euclidean similarity.

D Results on French datasets

Dataset	**Model**	**Layer**	$\ell < 500$		$\ell \geq 500$	
			$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
ontonotes	Mean of all models without fine-tuning		33.01	77.79	43.81	86.97
clres	Mean of all models without fine-tuning		43.42	76.48	51.29	78.99
eurosens_nouns	camembert-base	11	14.09	73.21	22.82	86.78
eurosens_verbs	camembert-base	11	6.302	64.66	7.555	67.96
eurosens_adjectives	camembert-base	11	15.92	69.48	19.01	83.65
eurosens_adverbs	camembert-base	11	12.31	74.72	19.24	78.66
eurosens_all	camembert-base	11	12.46	63.89	-	58.65
eurosens_nouns_swa	camembert-base	11	13.59	68.30	-	93.63
eurosens_verbs_swa	camembert-base	11	9.323	47.26	14.95	58.88
eurosens_adjectives_swa	camembert-base	11	13.65	56.09	-	68.69
eurosens_adverbs_swa	camembert-base	11	12.41	62.79	20.40	83.06
eurosens_all_swa	camembert-base	11	12.23	63.17	-	58.96
Mean eurosens	camembert-base	11	12.23	64.36	17.33	73.89

Table 12: **Mean Average Precision (MAP)** for Eurosens without fine-tuning.

The dataset "eurosens_all" means that it is a combined dataset containing nouns, verbs, adjectives and adverbs. The suffix "swa" refers to datasets for which we only retained only the single word annotations and suppressed the multi-word annotations.
A surprising result was that high-frequency, high-rarity words performed worse than low-frequency, high-rarity words in EuroSense. This pattern was not observed in PDEP or OntoNotes and may be due to subsampling biases, though further analysis was not possible due to computational constraints. Overall, EuroSense performed worse than PDEP and OntoNotes, likely due to the greater complexity of French, including rich morphology, flexible word order, and extensive grammatical agreements, which make word sense disambiguation more challenging. Finally, regardless of the number of runs performed, the high-frequency low-rarity bin was often empty, since frequent words tend to have highly specific labels. This was especially the case when multi-word expressions were removed, since despite being frequent, they had low rarity due to shared labels across multiple words, making disambiguation harder. We did not observe any significant difference in performance between single-word annotations and when multi-word annotations were kept

Dataset	**Model**	**Layer**	$\ell < 500$		$\ell \geq 500$	
			$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
ontonotes	Mean of all models without fine-tuning		9.3448	12.89	0.5781	0.2916
clres	Mean of all models without fine-tuning		6.4781	8.097	1.585	2.696
eurosens_nouns	camembert-base	11	1.790	9.407	0.9298	0.8214
eurosens_verbs	camembert-base	11	0.6539	8.159	0.1222	0.1386
eurosens_adjectives	camembert-base	11	1.893	7.422	0.4570	0.7113
eurosens_adverbs	camembert-base	11	1.356	9.326	0.4827	0.8775
eurosens_all	camembert-base	11	6.422	17.89	-	1.440
eurosens_nouns_swa	camembert-base	11	9.434	13.83	-	3.108
eurosens_verbs_swa	camembert-base	11	2.295	12.54	0.4741	0.3805
eurosens_adjectives_swa	camembert-base	11	3.659	8.786	-	1.720
eurosens_adverbs_swa	camembert-base	11	2.311	7.516	0.7192	0.3963
eurosens_all_swa	camembert-base	11	6.017	16.50	-	1.378
Mean eurosens	camembert-base	11	3.583	11.14	0.5308	1.097

Table 13: **Mean Recall** for EuroSens without fine-tuning.

Dataset	**Model**	**Layer**	$\ell < 500$		$\ell \geq 500$	
			$r < 0.25$	$r \geq 0.25$	$r < 0.25$	$r \geq 0.25$
ontonotes	Mean of all models without fine-tuning		13.55	21.18	1.136	0.5807
clres	Mean of all models without fine-tuning		10.80	14.22	3.108	5.158
eurosens_nouns	camembert-base	11	3.070	16.12	1.765	1.623
eurosens_verbs	camembert-base	11	1.151	14.01	0.2393	0.2764
eurosens_adjectives	camembert-base	11	3.273	13.03	0.8859	1.407
eurosens_adverbs	camembert-base	11	2.372	16.05	0.9344	1.730
eurosens_all	camembert-base	11	7.796	26.29	-	2.789
eurosens_nouns_swa	camembert-base	11	10.14	21.92	-	5.957
eurosens_verbs_swa	camembert-base	11	3.482	18.70	0.9104	0.7545
eurosens_adjectives_swa	camembert-base	11	5.446	14.60	-	3.330
eurosens_adverbs_swa	camembert-base	11	3.720	13.00	1.375	0.7876
eurosens_all_swa	camembert-base	11	7.443	24.68	-	2.673
Mean eurosens	camembert-base	11	4.789	17.84	1.018	2.133

Table 14: **Mean F1-score** for EuroSens without fine-tuning.

Dataset	**Model**	**Layer**	$\ell \in [1, 2[$	$\ell \in [2, 24[$
Flue	camembert-base	11	5.137	17.31

Table 15: **Mean Average Precision (MAP)** for Flue without fine-tuning.

Dataset	**Model**	**Layer**	$\ell \in [1, 2[$	$\ell \in [2, 24[$
Flue	camembert-base	11	89.48	79.58

Table 16: **Mean Recall** for Flue without fine-tuning.

Dataset	**Model**	**Layer**	$\ell \in [1, 2[$	$\ell \in [2, 24[$
Flue	camembert-base	11	9.128	25.36

Table 17: **Mean F1-score** for Flue without fine-tuning.